

Air purifier ideas

By Ayush Agarwal

<https://www.linkedin.com/in/ayush-agarwal-261041215/>

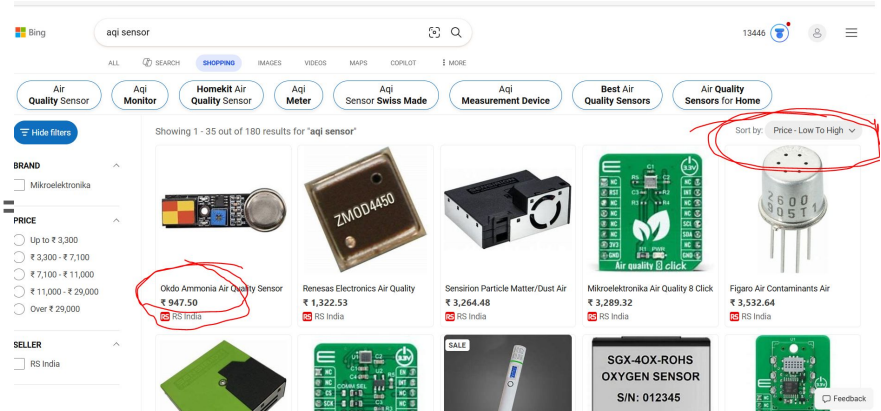
How to make it = First Principles Thinking

Lets break it into parts, and then solve.

ELIMINATABLE PARTS -

Parts that are “good to have” but not “must have” =

- AQI sensors (these are very expensive),
- Speed control for Fan circuit.



If you remove these both, you wont even need PCB, circuit will be simple.

PCB manufacturing + shipping is costly, and soldering the parts is labour intensive.

Dropping Chassis cost to near 0 !



Plastic Box -

- Costly.
- Not Sustainable
- Difficult to transport



Chart paper rolls -

- Ultra Cheap
- Sustainable
- Can be bulk transported and assembled in 1000s

[<https://www.linkedin.com/in/ayush-agarwal-261041215/>]]

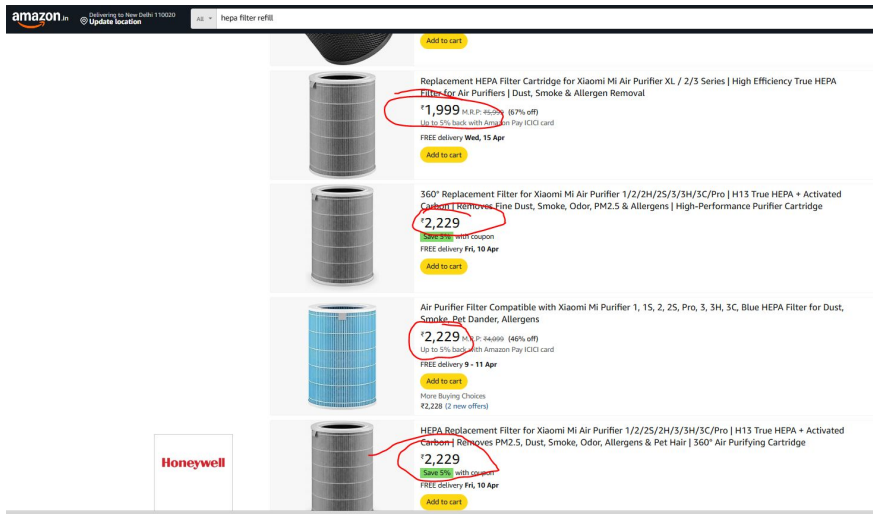
But how will chart paper even work -

Using origami skills, I believe it is possible.

[Proceeds to show **demo** with rough paper]

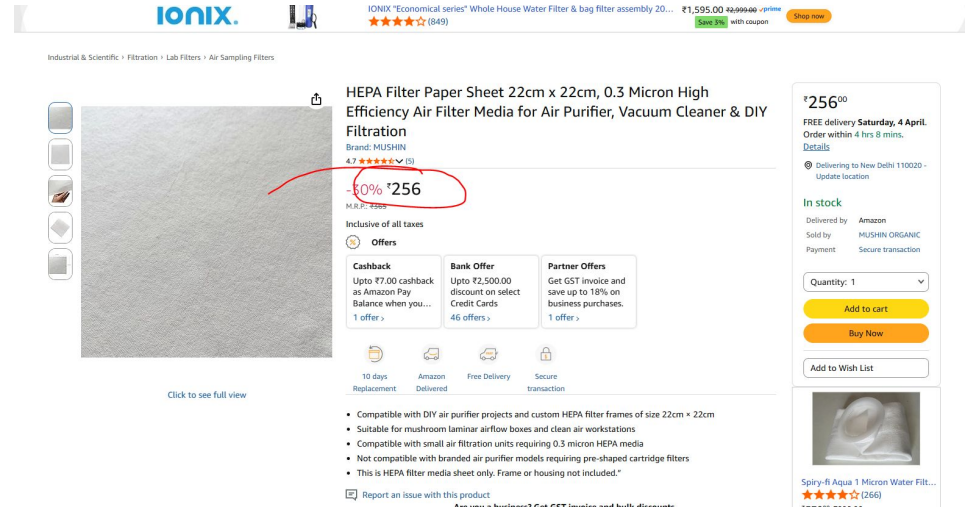
Note - Worst case, if it doesn't work, we can make rectangular pipe using Cardboard.

HEPA ?



Normal HEPA refills -

- Costs 2000 rupees on avg
- Paying mostly for compatibility with commercial purifiers

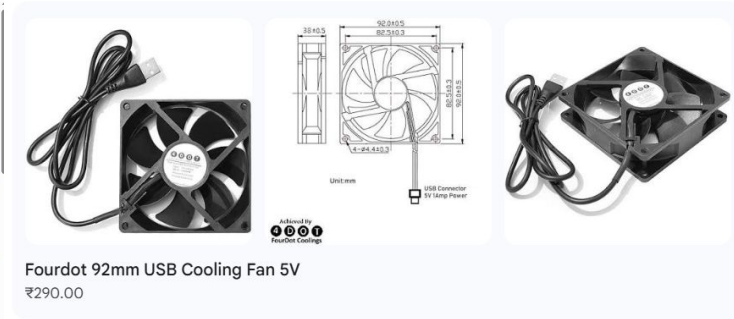
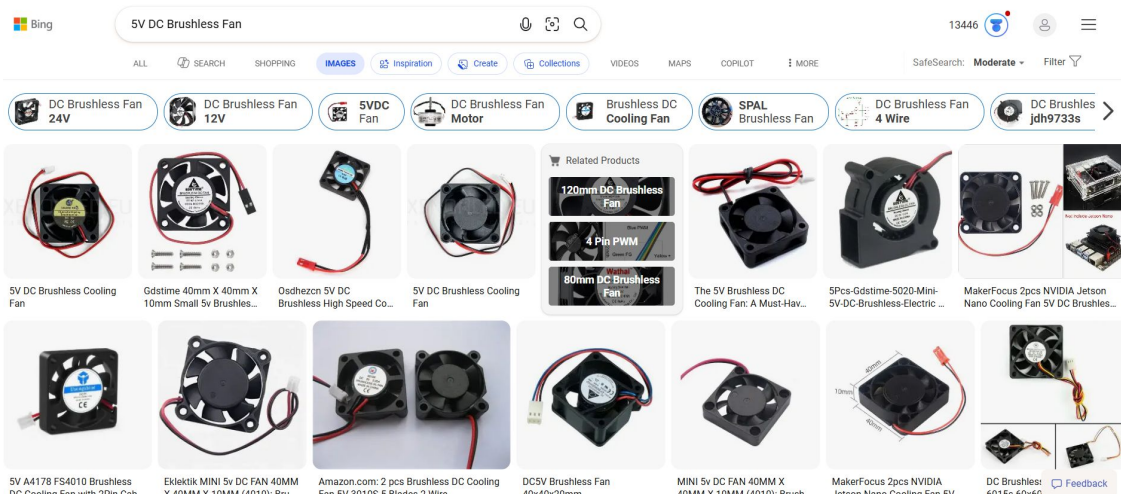


HEPA Sheets -

- Very cheap
- Cuttable into pieces

<https://www.linkedin.com/in/ayush-agarwal-261041215/>

Fans ?



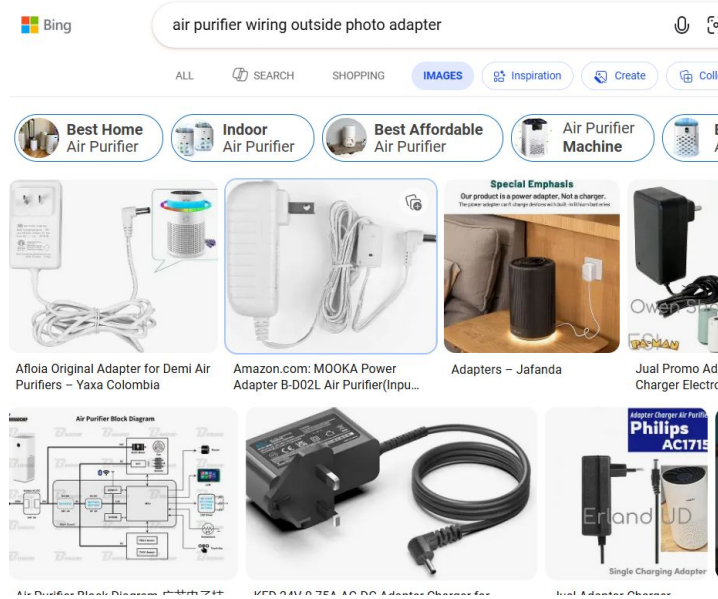
Remember these fans inside old CPUs?

We can use them, no need to manually attach propellor to motors etc

Drops manufacturing cost, plus no fire risk now

[<https://www.linkedin.com/in/ayush-agarwal-261041215/>]

Wiring ? Killing the need for adapter



Wiring/Adapter -

- Has 150-200 rupees cost

MUP



USB port attached to motor

Can make it compatible to phone chargers, thus removing need of sending the adapters.

[\[https://www.linkedin.com/in/ayush-agarwal-261041215/\]](https://www.linkedin.com/in/ayush-agarwal-261041215/)

Its a billboard

Since we are using chart paper.

We can sell the advertising spots on the chassis to companies, further dropping price by 20-30 rupees.

Plus CSR funding